

The no-three-collinear variant V3: a compact, search-free proof

Companion note

5 August 2026

Abstract

For $n \geq 4$, let $f_{V3}(n)$ be the least number of proper Euclidean circles determined by n distinct real planar points, no three collinear and not all concyclic. We prove

$$f_{V3}(8) = 20, \quad f_{V3}(n) = 1 + \binom{n-1}{2} \quad (n \neq 8).$$

The infinite range follows from a summed inverted Melchior inequality and two explicit lower bounds. The only equality obstructions left in the small range are proved here: a seven-point Fano wall and a ten-point six-circle power wall. No configuration, support, or orbit catalogue and no SAT, ILP, or finite-range search is used.

1 Statement and geometric conventions

Definition 1.1 (V3). A *V3 configuration* is a set $P \subset \mathbb{R}^2$ of $n \geq 4$ distinct points such that no three points of P are collinear and P is not contained in one proper circle. A circle is counted when it is a proper Euclidean circle containing at least three points of P , and it is counted only once, independently of the size of its full trace on P . We denote the count by $C(P)$ and its minimum by $f_{V3}(n)$.

Put

$$G(n) := 1 + \binom{n-1}{2}. \tag{1}$$

We use the following standard facts, always over the real plane. A noncollinear triple has a unique circumcircle; two distinct proper circles, or a proper circle and a line, meet in at most two points. Inversion about o interchanges the lines avoiding o with the proper circles through o , sends circles avoiding o to circles avoiding o , and preserves all incidences away from o . For a proper circle and two secants through a point, directed power of a point equates the two products of directed intercepts. Directed Menelaus applies to a nondegenerate real affine triangle and three collinear points on its extended sides; none of the six ratios used below has a zero denominator. The one global incidence input, Melchior's inequality, is derived in §3 for every finite noncollinear real projective point set. Thus all classical inputs are used on their stated real, finite, nondegenerate ranges.

Theorem 1.2. *For every integer $n \geq 4$,*

$$f_{V3}(n) = \begin{cases} 20, & n = 8, \\ G(n), & n \neq 8. \end{cases}$$

In particular,

$$(f_{V3}(4), \dots, f_{V3}(10)) = (4, 7, 11, 16, 20, 29, 37).$$

2 The two upper constructions

Construction 2.1 (A circle and one outsider). Choose $n - 1$ distinct points on a proper circle Γ , and choose $x \notin \Gamma$ outside every line through two of them. The forbidden set is a finite union of proper curves, so such an x exists. The resulting set has no collinear triple. It is not concyclic, since any circle through the $n - 1 \geq 3$ selected points of Γ equals Γ , while $x \notin \Gamma$. Thus it is V3. Its triples on Γ give Γ , and every other triple consists of x and a pair on Γ . Different pairs give different circles: a coincidence would produce a circle through x sharing at least three points with Γ . Hence

$$f_{V3}(n) \leq 1 + \binom{n-1}{2} = G(n). \quad (2)$$

The exceptional construction is just as explicit, but it is clearer before inversion, where lines and circles are temporarily treated together as *generalized circles*.

Construction 2.2 (The eight-point cross). Fix $0 < a < b$, let $S = \{-b, -a, a, b\}$, and set

$$Q = (S \times \{0\}) \cup (\{0\} \times S).$$

The two axes are four-point generalized circles. If $x_1 \neq x_2$ and $y_1 \neq y_2$, then

$$(x_1, 0), (x_2, 0), (0, y_1), (0, y_2) \text{ are concyclic} \iff x_1 x_2 = y_1 y_2. \quad (3)$$

Indeed, on restricting $X^2 + Y^2 + DX + EY + F = 0$ to the two axes, Vieta's formula says that both root pairs have product F ; conversely, equality of the products fixes D, E, F and gives a proper circle.

The six unordered pairs from S have four distinct product fibres,

product	ab	$-ab$	$-b^2$	$-a^2$
multiplicity	2	2	1	1.

Consequently (3) gives $2^2 + 2^2 + 1^2 + 1^2 = 10$ four-point $2 + 2$ circles. There are 48 mixed triples. Those ten circles own $10 \binom{4}{3} = 40$ of them. Each of the remaining eight triples has its own three-point circle: if two had the same carrier, that carrier could meet each axis at most twice, so their union would be a four-point $2 + 2$ support and hence one of the ten circles already counted. Thus the generalized blocks are

$$2 \text{ axes,} \quad 10 \text{ four-point circles,} \quad 8 \text{ three-point circles,}$$

and both the block and triple counts are exact:

$$2 + 10 + 8 = 20, \quad 2 \binom{4}{3} + 10 \binom{4}{3} + 8 = \binom{8}{3}. \quad (4)$$

Let $\mathcal{U}$ be the union of these twenty lines and circles. A finite union of such curves does not cover the plane, so choose $o \notin \mathcal{U}$ and invert about o . Since $Q \subset \mathcal{U}$, the centre is not a selected point. Every one of the twenty generalized blocks becomes a proper circle with the same selected trace. Conversely, a circle through an image triple pulls back to the unique generalized carrier of the original triple, so there are no additional blocks.

No three image points are collinear: the inverse of such a line would be a generalized circle containing o and an original triple, contrary to $o \notin \mathcal{U}$. The image is not concyclic either, since its inverse would put all of Q on one generalized circle, whereas a proper circle meets either axis at most twice and no line contains all eight points. Therefore

$$f_{V3}(8) \leq 20. \quad (5)$$

The product fibres, not a list of supports, explain the exception: after inversion the two rich axes and ten $2 + 2$ collisions compress all $\binom{8}{3}$ triples into only twenty proper circles.

3 Universal lower bounds

Fix a V3 configuration. Let c_s be the number of circles whose full selected trace has size s , let $C = \sum_s c_s$, and let $m = \max\{s : c_s > 0\}$. Unique circumcircles partition the selected triples, and nonconcyclicity gives $3 \leq m \leq n - 1$. Hence

$$\boxed{\mathsf{T} : \quad \sum_{s=3}^m \binom{s}{3} c_s = \binom{n}{3}}, \quad C = \sum_{s=3}^m c_s. \quad (6)$$

3.1 Summed inverted Melchior

For a finite noncollinear real projective point set, let t_r count its spanned lines containing exactly r selected points. Dualize to an essential arrangement of real projective lines. If V, E, F denote its numbers of vertices, edges, and faces, then

$$V = \sum_{r \geq 2} t_r, \quad E = \sum_{r \geq 2} r t_r, \quad V - E + F = 1.$$

Every face has at least three sides, so $3F \leq 2E$. Therefore $E \leq 3V - 3$, which is exactly

$$t_2 \geq 3 + \sum_{r \geq 4} (r - 3) t_r. \quad (7)$$

Invert now at a selected point p and remove p . The $n - 1$ images are noncollinear: a common image line through the centre would give a collinear original triple, while a common image line avoiding the centre would put all original points on one circle through p . Also, every spanned image line avoids the centre, again by the no-three-collinear hypothesis. Thus an s -point circle through p becomes an $(s - 1)$ -point line, with no exceptional line term.

Sum (7) over all n pivots. A three-circle contributes to t_2 at each of its three points; a four-circle has coefficient zero; and an s -circle, $s \geq 5$, contributes $s - 4$ at each of its s points. We obtain

$$\boxed{\mathsf{M} : \quad 3c_3 - \sum_{s=5}^m s(s - 4)c_s \geq 3n.} \quad (8)$$

3.2 The bounded-support functional

Assume $m \geq 4$, and put

$$A = \binom{m}{3}, \quad B = m(m - 4), \quad D = 3A + B, \quad \lambda_m = \frac{B + 3}{D}, \quad \mu_m = \frac{A - 1}{D}.$$

Here $D = m(m - 3)(m + 2)/2 > 0$, and both multipliers are positive. Add $\lambda_m \mathsf{T}$ and $\mu_m \mathsf{M}$. Directly from the definitions,

$$\lambda_m + 3\mu_m = 1, \quad \lambda_m \binom{m}{3} - \mu_m m(m - 4) = 1,$$

so the coefficients of c_3 and c_m are one. For every integer $4 \leq s \leq m$, the exact slack below one is

$$1 - \left(\lambda_m \binom{s}{3} - \mu_m s(s - 4) \right) = \frac{(m - s)(s - 3)(ms - m - s - 2)}{3m(m + 2)} \geq 0, \quad (9)$$

because $ms - m - s - 2 = (m - 1)(s - 1) - 3 \geq 6$. The weighted left side is therefore at most C , whereas its right side is a lower bound. Hence

$$C \geq L(n, m) := \lambda_m \binom{n}{3} + 3n\mu_m = \frac{n(3m^2 + mn^2 - 3mn + 2m - n^2 + 3n + 4)}{3m(m + 2)}. \quad (10)$$

Direct subtraction gives the useful strict monotonicity

$$L(n, m) - L(n, m + 1) = \frac{n(n - 4)(n + 1)(m^2 - m - 3)}{3m(m + 1)(m + 2)(m + 3)} > 0 \quad (11)$$

for $n \geq 5$ and $m \geq 4$.

3.3 The rich-circle pencil

Suppose a circle Γ contains m selected points, with $k = n - m$ outsiders. For each outsider x , the circles through x and a pair on Γ form a fan of exactly $\binom{m}{2}$ circles. A circle common to the fans of x and y uses the same pair on Γ . The pairs arising from two distinct common circles are disjoint, since otherwise the circles would share x, y and a point of Γ . Thus two fans meet in at most $\lfloor m/2 \rfloor$ circles. Two-term Bonferroni, together with Γ , yields

$$C \geq P(n, m) := 1 + (n - m) \binom{m}{2} - \binom{n - m}{2} \left\lfloor \frac{m}{2} \right\rfloor. \quad (12)$$

For later use suppose $m > \lfloor n/2 \rfloor$, so $1 \leq k < m$. If $m = 2r$, then $k \leq 2r - 1$ and

$$\begin{aligned} 2(P(n, 2r) - G(n)) &= (k - 1)(4r^2 - 6r + 2 - k(r + 1)), \\ 4r^2 - 6r + 2 - k(r + 1) &\geq (2r - 1)(r - 3). \end{aligned} \quad (13)$$

If $m = 2r + 1$, then $k \leq 2r$ and

$$\begin{aligned} 2(P(n, 2r + 1) - G(n)) &= (k - 1)(4r^2 - 2r - k(r + 1)), \\ 4r^2 - 2r - k(r + 1) &\geq 2r(r - 2). \end{aligned} \quad (14)$$

For $n \geq 11$, every relevant even or odd rich case has $r \geq 3$. Consequently

$$P(n, m) \geq G(n) \quad (n \geq 11, m > \lfloor n/2 \rfloor). \quad (15)$$

4 The whole infinite range

Let $n \geq 11$. If $m = 3$, then (6) gives $C = \binom{n}{3} > G(n)$. If $m > \lfloor n/2 \rfloor$, use (15). In the remaining range $4 \leq m \leq \lfloor n/2 \rfloor$, monotonicity gives

$$C \geq L(n, m) \geq L(n, \lfloor n/2 \rfloor).$$

For $n = 2r$, $r \geq 6$, exact subtraction yields

$$L(2r, r) - G(2r) = \frac{(r - 2)(2r^2 - 13r + 2)}{3(r + 2)}. \quad (16)$$

At $r = 6$ this is $-2/3$, and integrality of C and $G(12)$ rounds $C \geq G(12) - 2/3$ up to $C \geq G(12)$. At $r = 7$ the quadratic factor is 9, and thereafter its forward difference $4r - 11$ is positive.

For $n = 2r + 1$, $r \geq 5$, one obtains

$$L(2r + 1, r) - G(2r + 1) = \frac{(2r - 3)(r^3 - 4r^2 - 4r - 2)}{3r(r + 2)}. \quad (17)$$

The cubic factor is 3 at $r = 5$, and its forward difference $3r^2 - 5r - 7$ is positive for $r \geq 5$. The three ranges for m are disjoint and exhaustive, so

$$C \geq G(n) \quad (n \geq 11). \quad (18)$$

5 The two small real walls

5.1 Four-point circles and the Fano wall

When $m = 4$, (6) and (8) reduce to

$$\binom{n}{3} = c_3 + 4c_4, \quad c_3 \geq n. \quad (19)$$

Let $\rho(n)$ be the least integer at least n congruent to $\binom{n}{3} \pmod{4}$. Then

$$C = \frac{\binom{n}{3} + 3c_3}{4} \geq \frac{\binom{n}{3} + 3\rho(n)}{4}. \quad (20)$$

For $n = 5, \dots, 10$, the values of $\rho(n)$ are

$$(6, 8, 7, 8, 12, 12),$$

and (20) gives respectively

$$(7, 11, 14, 20, 30, 39). \quad (21)$$

Only the $n = 7$ equality lies below the eventual target.

Lemma 5.1 (The planar Fano wall). *Seven distinct real points cannot carry seven proper four-point circles whose three-point complements form a Steiner triple system STS(7).*

Proof. First normalize the Steiner system without a classification. Name one triple 123. The other triples through 1 pair the remaining four points, so relabel them as 145, 167. The triples through 2 must use the other pairings; after swapping 6, 7 if needed, take 246, 257. The two remaining unused pairs among 4, 5, 6, 7 are 47, 56, forcing

$$123, \quad 145, \quad 167, \quad 246, \quad 257, \quad 347, \quad 356. \quad (22)$$

Invert at point 1. The circles complementary to the last four triples in (22) become lines with full selected traces

$$357, \quad 346, \quad 256, \quad 247.$$

The first three have the three distinct pairwise intersections 3, 5, 6; they are therefore the sides of a nondegenerate triangle. Set

$$A = 3, \quad B = 5, \quad C = 6, \quad X = 7 \in AB, \quad Y = 4 \in AC, \quad Z = 2 \in BC.$$

The fourth line is the transversal XYZ . The three circles not through the inversion centre have traces

$$ABYZ, \quad ACXZ, \quad BCXY. \quad (23)$$

All vertices and marked points are distinct because the displayed traces are full traces of distinct carriers.

Use directed affine parameters

$$AB : (A, B, X) = (0, 1, x), \quad AC : (A, C, Y) = (0, 1, y), \quad BC : (B, C, Z) = (0, 1, z),$$

and put $a^2 = |BC|^2$, $b^2 = |CA|^2$, $c^2 = |AB|^2$. Directed power at C, B, A , respectively, for the circles in (23) gives

$$b^2(1 - y) = a^2(1 - z), \quad c^2(1 - x) = a^2z, \quad c^2x = b^2y.$$

The six factors $x, y, z, 1 - x, 1 - y, 1 - z$ are nonzero. Multiplying the three resulting ratios gives

$$\frac{x}{1-x} \frac{z}{1-z} \frac{1-y}{y} = +1. \quad (24)$$

But directed Menelaus for the transversal XYZ in triangle ABC gives the same product equal to -1 , a contradiction. $\square$

Return to $n = 7, m = 4$. Congruence in (19) says either $c_3 = 7$ or $c_3 \geq 11$. In the former case $c_4 = 7$. If A_i are the seven circle traces and $B_i = P \setminus A_i$, then for $i \neq j$

$$|B_i \cap B_j| = |A_i \cap A_j| - 1 \leq 1,$$

because two four-subsets of a seven-set meet and two distinct circles meet in at most two selected points. Thus the seven triples B_i contain $7 \binom{3}{2} = 21$ distinct pairs, exactly all pairs of P ; they form an STS(7), contrary to Lemma 5.1. Hence $c_3 \geq 11$, and (20) improves to

$$C \geq 17 \quad (n = 7, m = 4). \quad (25)$$

5.2 Six five-point circles on ten points

Lemma 5.2 (The six-five wall). *Ten distinct real points cannot support six distinct proper five-point circles if every two of their full traces meet in at most two points.*

Proof. Let the traces be $B_1, \dots, B_q$ in a ten-set X , and put $w_i = \mathbf{1}_{B_i} - \frac{1}{2}\mathbf{1}_X$. Then

$$\|w_i\|^2 = \frac{5}{2}, \quad \langle w_i, w_j \rangle = |B_i \cap B_j| - \frac{5}{2} \leq -\frac{1}{2}.$$

Therefore

$$0 \leq \left\| \sum_{i=1}^q w_i \right\|^2 \leq \frac{5q}{2} - \binom{q}{2} = \frac{q(6-q)}{2}. \quad (26)$$

In particular $q \leq 6$. If $q = 6$, equality throughout (26) forces

$$|B_i \cap B_j| = 2 \quad (i \neq j), \quad d(x) = 3 \quad (x \in X), \quad (27)$$

where the second conclusion follows coordinatewise from $\sum_i w_i = 0$.

We now derive, rather than assume, the equality design. Delete B_6 . Its five points have degree two among $B_1, \dots, B_5$, and the other five have degree three. Record the memberships of the first five as edges of a multigraph E on block labels $1, \dots, 5$; record the two-label complements of the memberships of the latter five as edges of a multigraph H . Each label has degree two in E , because $|B_i \cap B_6| = 2$. It also has degree two in H : among the five points outside B_6 , label i occurs three times and is omitted twice. Thus both graphs are 2-regular.

The graph E has no doubled edge. Otherwise two degree-two points would exhaust $B_i \cap B_j$, so no degree-three membership could contain both i, j . Yet i, j occur six times in total among the five degree-three memberships, impossible if each membership contains at most one of them. The graph H has no doubled edge either. If ij were doubled, the degrees of i, j would be exhausted and the other three vertices would form a triangle. Its three edges are disjoint from ij , so their complementary memberships would put at least three points in $B_i \cap B_j$, contradicting (27). Hence E, H are simple 2-regular graphs on five vertices, and thus both are five-cycles.

For a label pair ij , the number of H -edges disjoint from it is $1 + \mathbf{1}_{\{ij \in H\}}$. Counting $B_i \cap B_j$ now gives

$$2 = \mathbf{1}_{\{ij \in E\}} + 1 + \mathbf{1}_{\{ij \in H\}},$$

so E and H are complementary cycles. This proves uniqueness up to relabeling, without enumerating designs. Choose

$$E = \{14, 15, 23, 25, 34\}, \quad H = \{12, 13, 24, 35, 45\}.$$

The five memberships through block 6 are $6e$, $e \in E$, and the other five are the complements in $\{1, \dots, 5\}$ of the edges in H . Thus a forced normal form for the ten points is

$$123, 124, 135, 236, 146, 156, 245, 256, 345, 346. \quad (28)$$

This is one normal form derived from the two cycles, not a list of candidate supports.

Invert at the point 123. The three incident circles become distinct lines. Their pairwise second selected intersections 124, 135, 236 are distinct, so they form a nondegenerate triangle. Write

$$\begin{aligned} A &= 124, & B &= 135, & C &= 236, \\ X &= 146, & Y &= 156, & U &= 245, & V &= 256, & W &= 345, & Z &= 346. \end{aligned}$$

Then $X, Y \in AB$, $U, V \in AC$, and $W, Z \in BC$. The three nonincident circles have full traces

$$AXUWZ, \quad BYUVW, \quad CXYVZ. \quad (29)$$

Parametrize the sides by

$$\begin{aligned} AB : (A, B, X, Y) &= (0, 1, x, y), \\ AC : (A, C, U, V) &= (0, 1, u, v), \\ BC : (B, C, W, Z) &= (0, 1, w, z) \end{aligned}$$

and again put $a^2 = |BC|^2$, $b^2 = |CA|^2$, $c^2 = |AB|^2$. Power at the two omitted vertices for each circle in (29) gives all six identities

$$c^2(1-x) = a^2wz, \quad b^2(1-u) = a^2(1-w)(1-z), \quad (30)$$

$$c^2y = b^2uv, \quad a^2(1-w) = b^2(1-u)(1-v), \quad (31)$$

$$c^2xy = b^2v, \quad c^2(1-x)(1-y) = a^2z. \quad (32)$$

The nine image points are distinct, so each of

$$x, y, u, v, w, z, \quad 1-x, 1-y, 1-u, 1-v, 1-w, 1-z$$

is nonzero; every following division is legal. Dividing the fifth identity by the third, the sixth by the first, and combining the second with the fourth gives

$$x = \frac{1}{u}, \quad 1-y = \frac{1}{w}, \quad (1-v)(1-z) = 1. \quad (33)$$

The first, fourth, and third identities, with their squared side lengths cancelled, also give the coefficient-free cycle

$$\frac{(1-x)(1-w)uv}{wz(1-u)(1-v)y} = 1. \quad (34)$$

Substitution from (33) makes the left side of (34) equal to

$$\frac{((u-1)/u)(1-w)uv}{w(v/(v-1))(1-u)(1-v)((w-1)/w)} = -1.$$

Every displayed factor is nonzero by the preceding guard. This contradiction proves that $q = 6$ is impossible over $\mathbb{R}$. $\square$

Now take $n = 10, m = 5$ and write $q = c_5$. Circle intersection gives the hypothesis of Lemma 5.2, so $q \leq 5$. Equations (6) and (8) become

$$120 = c_3 + 4c_4 + 10q, \quad 3c_3 - 5q \geq 30. \quad (35)$$

Modulo four, $c_3 \equiv -2q \pmod{4}$. For $0 \leq q \leq 5$, every candidate through $8 + 2q$ violates the second inequality, since $3(8 + 2q) - 5q = 24 + q < 30$; all smaller integers in the same residue class do as well. The next congruent integer is $12 + 2q$, so $c_3 \geq 12 + 2q$, and elimination of c_4 gives

$$C = 30 + \frac{3}{4}c_3 - \frac{3}{2}q \geq 39 \quad (n = 10, m = 5). \quad (36)$$

6 The fixed small dispatch and conclusion

For $4 \leq n \leq 10$, every possible largest trace size is one of $m = 3, \dots, n - 1$. In that order, the lower bounds obtained from (6), (21), (25), (36), and (12) are

$$\begin{aligned} n = 4 : & \quad (4), \\ n = 5 : & \quad (10, 7), \\ n = 6 : & \quad (20, 11, 11), \\ n = 7 : & \quad (35, 17, 19, 16), \\ n = 8 : & \quad (56, 20, 25, 28, 22), \\ n = 9 : & \quad (84, 30, 29, 37, 40, 29), \\ n = 10 : & \quad (120, 39, 39, 43, 55, 53, 37). \end{aligned}$$

This is a seven-line substitution into proved formulas, not a catalogue of configurations or supports. Taking the minimum in each line gives

$$4, 7, 11, 16, 20, 29, 37.$$

Construction 2.1 attains every value except at eight; Construction 2.2 attains twenty at eight. Together with (18), this proves Theorem 1.2.

Computational boundary

No computation is a premise of this proof. Exact-arithmetic companion programs can corroborate the polynomial factorizations (9)–(17), the two product-fibre identities, the forced incidence normal forms, and the final sign cancellations. They do not formalize projective Euler topology, inversion, directed power, or Menelaus, and they do not establish any geometric incidence used above. Conversely, this text uses no finite scan in n or m , no generated configuration or support family, no orbit reduction, and no solver output.